Introduction to Probability and Statistics

Exercises Lecture 2

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Exercise 1: Repeat the last exercise of the lecture with different parameters: There's a bag with 3 balls. Each ball is either orange or green. The number of green balls is $\theta \in \{0, 1, 2, 3\}$. Estimate θ by grabbing and putting back a ball 5 times for the cases:

a) 4 out of the 5 of the balls are green. Calculate the likelihood for each possible value of $\theta \in \mathbb{N}$ (i.e., θ must be an integer).
b) 4 out of the 5 of the balls are green. Calculate the likelihood and maximize the likelihood function for $\theta \in \mathbb{R}$ and then choose the best valid integer.
c) 3 out of the 5 of the balls are green. Calculate the likelihood for each possible value of $\theta \in \mathbb{N}$ (i.e., θ must be an integer).
d) 3 out of the 5 of the balls are green. Calculate the likelihood and maximize the likelihood function for $\theta \in \mathbb{R}$ and then choose the best valid integer.

Exercise 2: The room temperature during a PhD course was recorded using a Raspberry Pi and a temperature sensor. The collected values are in the file temperature.csv (the temperature is the second column).

a) Calculate the sample mean $\overline{X}_n$ and variance S_n^2 of the temperature with all the $n = 253$ measurements
b) Assume that $\overline{X}_n$ is the true temperature μ and that S_n^2 is the real variance of the Gaussian noise that affects each measurement of the sensor, denoted by σ^2. Consequently, assume that the temperature measurements have a distribution $X_i \sim \mathcal{N}(\mu, \sigma^2)$. Plot the likelihood or log-likelihood function for the first 10 and with the first 100 measurements
c) Calculate the Maximum Likelihood Estimator (MLE) for μ with the first 10 and with the first 100 measurements under the previous assumption.